Figaro: Self-Enforcing Multi-Party Coordination via Asymmetric Bonding

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Abstract

We present Figaro, a smart-contract protocol that enables self-enforcing agreements between mutually distrusting parties without arbitrators, timeouts, or governance mechanisms. The core primitive is *asymmetric bonding*: each party locks collateral equal to twice the transaction value; only the buyer can trigger resolution. Cooperation is thereby the strictly dominant strategy in a one-shot game, producing a Nash equilibrium that holds without repeated interaction.

The mechanism scales from two-party exchange to N-party process trees through *progressive collateralization*: each seller bonds against the cumulative value of all upstream work, not merely their own payment. We prove that this preserves the Nash equilibrium at every chain position, with the risk-to-reward ratio increasing geometrically for downstream participants.

All orders within a process resolve *atomically*—all or nothing—creating endogenous coordination pressure among sellers analogous to group lending in microfinance. This yields a three-layer enforcement architecture: (1) economic self-interest via bonding, (2) inter-seller coordination pressure via atomic resolution, and (3) legal deterrence backed by immutable on-chain evidence.

We describe the mechanism formally, prove its game-theoretic properties, present its implementation as a Solidity smart contract verified by TLA$^+$ model checking (7 invariants, 2M+ states) and property-based fuzzing, and discuss implications for organizational theory—specifically, the Coasean thesis that firms exist to minimize transaction costs, arguing that sufficiently low enforcement cost shifts coordination toward transaction-scoped institutions assembled directly from bonded counterparties.

Keywords: mechanism design, Nash equilibrium, smart contracts, collateral bonding, multi-party coordination, process trees, post-firm economy

1 Introduction

1.1 The Problem

Economic coordination between strangers requires trust infrastructure. When two parties who have never met wish to exchange value, they face the classic commitment problem: either party may defect after the other has performed. The historical solution is *institutional intermediation*—firms, courts, escrow services, platform operators—each of which introduces costs, delays, jurisdictional dependencies, and power asymmetries.

The problem intensifies in multi-party settings. A food delivery involves a cook, a kitchen operator, an ingredient sourcer, and a courier; a procurement process involves engineers, manufacturers, inspectors, and logistics providers. Traditional coordination architectures address this complexity by aggregating participants into *firms*—legal entities that internalize transaction

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costs at the expense of individual sovereignty. Coase [Coase, 1937] formalized this observation: firms exist because the cost of discovering prices, negotiating terms, and enforcing contracts in open markets exceeds the cost of hierarchical management.

Blockchain-based smart contracts have produced a rich literature on trustless exchange, but most protocols address financial operations—swaps, lending, derivatives—rather than general coordination. Those that target coordination (e.g., Kleros 2019, Aragon 2017) typically reintroduce the intermediary as an on-chain dispute resolution mechanism: a jury, a governance body, or a timeout that unilaterally releases funds.

1.2 Our Contribution

We present FIGARO, which achieves self-enforcing coordination through a single mechanism: *asymmetric bonding.* The protocol's contributions are:

(i) **A minimal settlement primitive** with two external functions, three mappings, no owner, no fee, and no escape hatches from the bonded state. We show that this minimality is not a simplification but a *security requirement*: every additional exit path weakens the Nash equilibrium.

(ii) **Progressive collateralization** for N-party process trees. We prove (Theorem 5.3) that cooperation remains strictly dominant at every position in an arbitrary-depth process tree, with the seller's risk-to-reward ratio increasing with chain depth.

(iii) **Atomic resolution** as a coordination forcing function. We formalize the analogy between all-or-nothing process settlement and group lending in microfinance, showing that atomic resolution creates endogenous peer pressure among sellers without explicit communication or governance.

(iv) **Self-enforcing DAG topology** without on-chain validation. We prove (Theorem 5.10) that misreporting cumulative value is a strictly dominated strategy, making on-chain parent-relationship validation unnecessary.

(v) **Formal verification.** The implementation is verified through TLA$^+$ model checking (7 invariants, 2M+ states exhaustively explored), property-based fuzzing (7 Echidna invariants, 50,000 call sequences), and 180 Foundry unit tests.

(vi) **Implications for organizational theory.** We argue that asymmetric bonding prices the cost of trust at exactly $2\times$ the transaction value, potentially replacing the standing firm with transaction-scoped institutions assembled directly around each process.

1.3 Paper Organization

Section 2 surveys related work. Section 3 presents the mechanism formally. Section 4 proves the game-theoretic properties. Section 5 extends the model to N-party process trees and DAGs. Section 6 describes the smart contract implementation and verification. Section 7 analyzes attack vectors. Section 8 discusses implications for organizational economics. Section 9 concludes with open questions and future work.

2 Related Work

Mutual-destruction escrows. The direct ancestor of FIGARO is the Safe Remote Purchase contract [Solidity Team, 2016], included in the Solidity documentation as a reference pattern: buyer and seller each lock $2\times$ the item value, and only mutual agreement releases the funds.

This two-party, single-order pattern is the seed from which FIGARO grew. BitHalo [BitHalo, 2014] independently explored mutual-destruction deposits for peer-to-peer Bitcoin trade.

The central problem was generalizing SRP beyond two parties. SRP uses symmetric 2× deposits with mutual agreement to release—both parties must cooperate. This works for a single exchange but cannot compose: in a twelve-seller delivery workflow, there is no natural "mutual agreement" among all participants. *Asymmetric bonding* (Theorem 3.1) solves this. By scaling each seller's bond with cumulative upstream value rather than fixing it per-trade, the mechanism creates a mesh of independently secured edges. Each edge carries its own Nash equilibrium (Theorem 4.3), and the mesh decomposes as finely as gas costs permit.

Buyer dominance then takes advantage of this mesh structure to make the standing firm non-compulsory. Because a single party resolves the entire process tree atomically, coordination pressure propagates through the mesh without any management layer (Theorem 5.8). What was previously internalized inside a firm—scheduling, quality control, supplier management—is now enforced by the bonding game itself. The organizational unit becomes process-local and temporary rather than a standing managerial container.

Extending to N parties followed naturally once buyer dominance was established—the cumulative value accumulator and atomic resolution are direct consequences. The difficult work was *reduction*: removing timeouts, owner functions, governance mechanisms, internal ledgers, withdrawal patterns, position NFTs, and every other feature that development pressure—from collaborators, AI agents, and inherited DeFi/web2 mental models—insisted on adding. The result is a minimal kernel: two external functions, three mappings, no owner, no fee, no escape hatches. Each removed feature was not merely unnecessary but actively harmful—violating one of the six properties derived in Section 4.

State channels and payment networks. The Lightning Network [Poon and Dryja, 2016] and Raiden [Raiden, 2017] use time-locked deposits for off-chain payment. These protocols optimize for high-frequency bilateral payments and rely on timeouts for dispute resolution—precisely the escape hatch that FIGARO eliminates. More critically, state channels do not address multi-party *coordination* (delivery, fulfillment, service composition); they address multi-hop *payment routing*.

On-chain dispute resolution. Kleros [Kleros, 2019] provides decentralized arbitration via Schelling-point juror selection. Aragon Court [Aragon, 2017] offers similar functionality within DAO governance. Both reintroduce a *discretionary human decision* into the resolution path, breaking the self-enforcing property. In FIGARO, disputes are resolved by economic pressure (Layer 1), peer coordination (Layer 2), and—only in extremis—off-chain legal proceedings backed by immutable on-chain evidence (Layer 3). No on-chain governance mechanism exists.

Optimistic mechanisms. Optimistic rollups [Optimism, 2021] and optimistic oracles [UMA, 2020] use a challenge-response pattern: an assertion stands unless challenged within a timeout window. This requires liveness assumptions (the challenger must be online) and timeout calibration (too short enables fraud; too long delays finality). FIGARO has no timeout from the bonded state—capital remains locked indefinitely until the buyer resolves, making liveness irrelevant to security.

Mechanism design. The revelation principle [Myerson, 1981] guarantees that any achievable outcome can be implemented via a direct mechanism. Figaro's insight is that the "direct mechanism" for multi-party coordination need not involve message passing, reporting, or arbitration at all: locked capital *is* the mechanism. The closest theoretical antecedent is the concept of *deposit games* [Asgaonkar and Krishnamachari, 2019], which analyze two-party security deposits. We extend deposit game analysis to N-party DAGs with progressive collateralization.

Transaction cost economics. Coase [Coase, 1937] argued that firms exist to minimize the transaction costs of market exchange. Williamson [Williamson, 1985] refined this into Transaction Cost Economics (TCE), identifying asset specificity, uncertainty, and frequency as drivers of hierarchical governance. We argue (Section 8) that asymmetric bonding provides a *fixed-cost* alternative to these institutional solutions, potentially altering the make-or-buy boundary that defines the firm.

3 The Mechanism

3.1 Notation and Setup

Consider two parties, a *buyer* B and a *seller* S, who wish to exchange a payment $P > 0$ (denominated in an ERC-20 token) for goods or services. Both parties hold sufficient token balances.

Definition 3.1 (Asymmetric Bonding)**.** An *asymmetric bond* for a transaction with payment P and cumulative value $G \geq P$ is a pair of deposits:

$$C_b = 2P \quad \text{(buyer bond)}, \tag{1}$$
$$C_s = 2G \quad \text{(seller bond)}. \tag{2}$$

For a root transaction (no upstream value), $G = P$, so both bonds equal $2P$. For sub-orders in a process tree, $G > P$, creating an asymmetry where the seller's bond exceeds the buyer's.

Definition 3.2 (Commitment)**.** A *commitment* is a tuple $\langle B, S, \tau, P, h, \sigma, \delta \rangle$ where B is the buyer address, S the seller address, τ the token contract, P the payment, h a content hash of the off-chain agreement, σ a cryptographic salt, and δ a signature expiry deadline. Both parties sign the commitment off-chain using EIP-712 typed structured data [EIP-712, 2017].

Definition 3.3 (Process)**.** A *process* Π is a tuple $\langle \text{id}, B, \tau, G, n, \mathcal{O} \rangle$ where id is a deterministic identifier derived from the root commitment, B is the root buyer (constant across all orders), τ is the denomination token (constant), G is a *monotonic accumulator* tracking total committed value, n is the count of active orders, and $\mathcal{O}$ is the set of active order identifiers.

3.2 Protocol Operations

The protocol exposes exactly two state-changing operations.

Operation 1: `commit`. Given a commitment c signed by both B and S:

1. Verify both EIP-712 signatures.
2. If $c.\text{processId} = 0$, derive a new process identifier from the typed-data digest and initialize the process with $G \leftarrow P$ and $n \leftarrow 1$.
3. Otherwise verify that $c.\text{processId}$ identifies an existing process, that signer B matches the process root buyer (*buyer dominance*), and that $c.\text{expectedCumulativeValue} = \Pi.G + P$.
4. Pull $C_b = 2P$ from B and $C_s = 2c.\text{expectedCumulativeValue}$ from S.
5. Create or extend the process accordingly, update the monotonic accumulator, and emit an evidence event that fully records the commitment.

Operation 2: `resolveProcess`**.** Given a process identifier and the complete list of active commitments:

1. Verify `msg.sender` $= \Pi.B$ (buyer dominance).
2. Verify that the list length equals $\Pi.n$ (atomic resolution: *all* orders must be included).
3. For each order o_i with payment P_i and cumulative snapshot G_i:

$$\pi_{s,i} = 2G_i + P_i, \tag{3}$$
$$\pi_{b,i} = P_i. \tag{4}$$

4. Transfer $\pi_{s,i}$ to each seller, $\pi_{b,i}$ to the buyer.
5. Mark all orders as resolved.

Remark 3.4 (Conservation Law)*.* For each order, the total distributed equals total locked:

$$\pi_{s,i} + \pi_{b,i} = (2G_i + P_i) + P_i = 2G_i + 2P_i = C_{s,i} + C_{b,i}. \tag{5}$$

The contract holds zero balance after resolution.

Remark 3.5 (No Escape Hatches)*.* There is no timeout, no cancel-after-commitment, no admin override, no governance vote, and no unilateral exit from the bonded state. Once both parties have committed, the only path to fund recovery is buyer-initiated resolution. This is not a temporary simplification; it is a *security requirement.* Theorem 4.7 proves that any additional exit path weakens the Nash equilibrium.

3.3 Net Payoffs

Definition 3.6 (Generalized Payout Functions)**.** For a bond multiplier $k \geq 1$, an order with payment $P > 0$ and cumulative value $G \geq P$, define the *settlement payout functions*:

$$\pi_s(k, G, P) = k \cdot G + P, \tag{6}$$
$$\pi_b(k, P) = (k - 1) \cdot P. \tag{7}$$

These satisfy the conservation law $\pi_s + \pi_b = k \cdot G + k \cdot P = C_s + C_b$ for $C_s = k \cdot G$ and $C_b = k \cdot P$. The *net payoff* is the payout minus the deposited bond:

$$u_S^{\text{net}}(k) = \pi_s - C_s = P, \tag{8}$$
$$u_B^{\text{net}}(k) = \pi_b - C_b = -P. \tag{9}$$

Remark 3.7*.* For the protocol's $k = 2$: $\pi_s = 2G + P$, $\pi_b = P$, $C_s = 2G$, $C_b = 2P$. The net payoffs ($+P$ for the seller, $-P$ for the buyer) are independent of k—the multiplier affects only the capital at risk, not the final value transfer.

4 Game-Theoretic Analysis

4.1 Two-Party Nash Equilibrium

Definition 4.1 (The Bonding Game)**.** The *bonding game* $\Gamma = (\{B, S\}, \{A_B, A_S\}, \{u_B, u_S\})$ is a one-shot, simultaneous-move, complete-information game:

- **Players**: buyer B, seller S.
- **Action sets**: $A_B = A_S = \{\mathrm{C}, \mathrm{D}\}$.

- **Interpretation**: C for S means delivering the agreed goods/services; C for B means calling `resolveProcess`. D means failing to perform.
- **Payoff functions** (net of bond deposits), for a root order with payment $P > 0$ and $G = P$:

$$u_B(a_B, a_S) = \begin{cases} -P & \text{if } a_B = \mathrm{C} \text{ and } a_S = \mathrm{C}, \\ -2P & \text{otherwise.} \end{cases} \tag{10}$$

$$u_S(a_B, a_S) = \begin{cases} +P & \text{if } a_B = \mathrm{C} \text{ and } a_S = \mathrm{C}, \\ -2G & \text{otherwise.} \end{cases} \tag{11}$$

The "otherwise" payoff is identical across all non-cooperative outcomes because any defection prevents resolution: bonds remain locked indefinitely, yielding a net loss equal to the full deposit.

Table 1: Payoff matrix for the bonding game Γ. Entry format: (u_B, u_S). Parameters: $P > 0$, $G \geq P > 0$.

	$a_B = \mathrm{C}$	$a_B = \mathrm{D}$
$a_S = \mathrm{C}$	$(-P,\ +P)$	$(-2P,\ -2G)$
$a_S = \mathrm{D}$	$(-2P,\ -2G)$	$(-2P,\ -2G)$

Definition 4.2 (Dominance)**.** Action a_i *weakly dominates* a_i' for player i if $u_i(a_i, a_{-i}) \geq u_i(a_i', a_{-i})$ for all a_{-i}, with strict inequality for at least one a_{-i}. If the inequality is strict for *all* a_{-i}, the dominance is *strict*.

Theorem 4.3 (Two-Party Nash Equilibrium)**.** *In the bonding game* Γ *of Theorem 4.1:*

(a) C *weakly dominates* D *for both the buyer and the seller.*

(b) (C, C) *is the unique Nash equilibrium.*

(c) (C, C) *is the unique Pareto-optimal outcome.*

Proof. **Part (a): Weak dominance.**

Buyer. We compare $u_B(\mathrm{C}, a_S)$ vs. $u_B(\mathrm{D}, a_S)$ for each seller action:

$$a_S = \mathrm{C}: \quad u_B(\mathrm{C}, \mathrm{C}) = -P > -2P = u_B(\mathrm{D}, \mathrm{C}), \tag{12}$$

$$a_S = \mathrm{D}: \quad u_B(\mathrm{C}, \mathrm{D}) = -2P = u_B(\mathrm{D}, \mathrm{D}). \tag{13}$$

Inequality (12) is strict (since $P > 0$); (13) is an equality. By Theorem 4.2, C weakly dominates D for B.

Seller. We compare $u_S(a_B, \mathrm{C})$ vs. $u_S(a_B, \mathrm{D})$ for each buyer action:

$$a_B = \mathrm{C}: \quad u_S(\mathrm{C}, \mathrm{C}) = +P > -2G = u_S(\mathrm{C}, \mathrm{D}), \tag{14}$$

$$a_B = \mathrm{D}: \quad u_S(\mathrm{D}, \mathrm{C}) = -2G = u_S(\mathrm{D}, \mathrm{D}). \tag{15}$$

Inequality (14) is strict (since $P > 0$ and $G > 0$ imply $P + 2G > 0$); (15) is an equality. C weakly dominates D for S.

Part (b): Unique Nash equilibrium. Since C weakly dominates D for both players, neither can profitably deviate from (C, C). We verify no other profile is a Nash equilibrium:

- (C, D): Seller deviates to C, gaining $P - (-2G) = P + 2G > 0$. Not a NE.
- (D, C): Buyer deviates to C, gaining $-P - (-2P) = P > 0$. Not a NE.

- (D, D): Buyer deviates to C with no change (both yield $-2P$). Seller deviates to C with no change (both yield $-2G$). This profile satisfies the Nash condition (no *profitable* deviation), but an iterated elimination of weakly dominated strategies removes D for both players, leaving (C, C) as the unique survivor.

Part (c): Pareto optimality. The payoff vector $(-P, +P)$ Pareto-dominates $(-2P, -2G)$ since $-P > -2P$ and $+P > -2G$. □

Remark 4.4 (On Weak vs. Strict Dominance). The dominance is weak, not strict, because when the counterparty defects, both actions yield the same locked-capital payoff. This is intrinsic to the mechanism: once one party defects, the outcome is fully determined regardless of the other's choice. The critical property is that (D, D) does not survive iterated elimination of weakly dominated strategies—the standard refinement for mechanism design [Myerson, 1981].

4.2 Sufficiency of the $2\times$ Multiplier

We now characterize the multiplier k in the generalized bonding game where $C_s = k \cdot G$, $C_b = k \cdot P$, and payouts are given by Theorem 3.6.

Lemma 4.5 (Buyer Indifference at $k = 1$). *For $k = 1$, the buyer is indifferent between cooperating and defecting when the seller cooperates: $u_B(\mathrm{C}, \mathrm{C}) = u_B(\mathrm{D}, \mathrm{C}) = -P$.*

Proof. With $k = 1$: $C_b = P$ and $\pi_b = (k-1)P = 0$ (from (7)). Under our accounting convention (Theorem 3.6), the buyer's net payoff under cooperation is $u_B(\mathrm{C}, \mathrm{C}) = \pi_b - C_b = 0 - P = -P$. Under defection, the buyer's bond remains locked: $u_B(\mathrm{D}, \mathrm{C}) = -C_b = -P$. Since $u_B(\mathrm{C}, \mathrm{C}) = u_B(\mathrm{D}, \mathrm{C}) = -P$, the buyer has no strictly profitable deviation from D—the mechanism provides zero marginal incentive to resolve. □

Theorem 4.6 (Minimal Viable Bond Multiplier). *Let $k \in \mathbb{Z}_{\geq 1}$ be the bond multiplier. Then:*

(a) For $k = 1$, C *does not weakly dominate* D *for the buyer: the buyer is indifferent when the seller cooperates.*

(b) For $k = 2$, C *weakly dominates* D *for both players (Theorem 4.3).*

(c) For $k > 2$, the equilibrium properties are identical to $k = 2$, but capital requirements increase by $(k-2)(G + P)$ per order with no improvement in dominance relations.

Thus $k = 2$ is the minimal integer multiplier that yields weak dominance of C *for both players, and is Pareto-optimal in capital efficiency among all valid multipliers.*

Proof. **Part (a).** See Theorem 4.5. With $k = 1$, the buyer recovers $\pi_b = 0$ upon resolution, having deposited $C_b = P$. The net monetary payoff from cooperating is $-P$; the net from defecting is also $-P$ (bond locked). The buyer cannot be motivated by the mechanism alone to resolve.

Part (b). With $k = 2$: $\pi_b = P$, $C_b = 2P$. Cooperation yields $u_B = -P$; defection yields $u_B = -2P$. Since $-P > -2P$ (as $P > 0$), the buyer strictly prefers cooperation when the seller cooperates. By Theorem 4.3(a), this establishes weak dominance.

Part (c). For $k \geq 2$: $u_B(\mathrm{C}, \mathrm{C}) = -P$ and $u_B(\mathrm{D}, \mathrm{C}) = -kP$. Since $-P > -kP$ for all $k \geq 2$, the dominance relation holds. The surplus capital locked, $(k-2)P$ for the buyer and $(k-2)G$ for the seller, earns zero return and is pure deadweight cost. □

4.3 Escape-Hatch Weakness

Theorem 4.7 (Escape-Hatch Weakness)**.** *Let Γ_E be the bonding game augmented with a unilateral exit action E available to player i, returning fraction $\alpha \in (0, 1]$ of player i's bond. Then either:*

(a) C *no longer weakly dominates* D *for at least one player (the Nash equilibrium of Theorem 4.3 may not survive), or*

(b) E requires a decision by a third party $J \notin \{B, S\}$ whose incentives are not constrained by the bond structure.

Proof. We analyze three exhaustive sub-cases of unilateral exit.

Case 1: Timeout (automatic exit after duration T). The action space becomes $A_B = \{\mathrm{C}, \mathrm{D}, \mathrm{Wait}\}$ where Wait means the buyer does not resolve and collects $\alpha \cdot C_b$ after time T. The buyer's payoff under Wait against a cooperating seller is:

$$u_B(\mathrm{Wait}, \mathrm{C}) = -C_b + \alpha \cdot C_b = -2P(1 - \alpha). \tag{16}$$

Compare with cooperation: $u_B(\mathrm{C}, \mathrm{C}) = -P$. The buyer prefers C to Wait iff:

$$-P > -2P(1 - \alpha) \iff 2(1 - \alpha) > 1 \iff \alpha < \tfrac{1}{2}. \tag{17}$$

For $\alpha \geq \frac{1}{2}$, Wait yields $u_B \geq -P = u_B(\mathrm{C}, \mathrm{C})$, so C no longer (weakly) dominates Wait. The buyer may rationally wait out the timeout instead of resolving. Even for $\alpha < \frac{1}{2}$, the defection cost drops from $2P$ to $2P(1-\alpha)$, weakening the deterrent. The seller, anticipating the timeout, faces a modified game where the buyer's commitment to resolve is no longer credible—the timeout provides an alternative path to partial capital recovery, undermining the "no escape" property that sustains the equilibrium.

Case 2: Arbitrator override. A third party J can invoke E on behalf of either player. Formally, this extends the game to three players $\{B, S, J\}$. The mechanism's self-enforcing property requires that the payoff structure *alone* determines the outcome. Introducing J makes the outcome contingent on J's action, and J is not bonded: J deposits nothing, risks nothing, and has no mechanism-constrained incentive to decide correctly. Whether J is an individual (arbitrator), an algorithm (oracle), or a collective (jury), the resolution path now depends on an agent outside the bond structure. This is condition (b).

Case 3: Governance vote. A special case of Case 2 where J is replaced by a set of voters $\{J_1, \ldots, J_m\}$ who collectively decide on E. This inherits all problems of Case 2, plus: (i) coordination failures among voters, (ii) plutocratic capture if voting power is token-weighted, (iii) voter apathy (rational ignorance). Each voter J_k is unbonded.

In all three cases, either C ceases to weakly dominate all alternatives (Case 1), or resolution depends on an unbonded external agent (Cases 2–3). □

Corollary 4.8. *No timeout duration T and no recovery fraction $\alpha > 0$ can be added to the bonding game without weakening the buyer's incentive to cooperate. In particular, full recovery ($\alpha = 1$) reduces the game to a zero-cost option, making defection strictly dominant for the buyer.*

5 N-Party Process Trees

5.1 Progressive Collateralization

We now extend the two-party model to an arbitrary-length chain of sellers.

Definition 5.1 (Process Tree)**.** A *process tree* is a sequence of bonded orders $\langle o_1, o_2, \ldots, o_n \rangle$ sharing a common buyer B and token τ. Order o_i has local payment $P_i > 0$ and cumulative value:

$$G_i = \sum_{j=1}^{i} P_j. \tag{18}$$

The seller of o_i posts bond $C_{s,i} = 2G_i$; the buyer posts bond $C_{b,i} = 2P_i$.

Definition 5.2 (N-Party Bonding Game)**.** The N*-party bonding game* $\Gamma_N = (\{B, S_1, \ldots, S_n\}, \{A_i\}, \{u_i\})$ extends Γ to n sellers. Each seller S_i has action set $\{\mathrm{C}, \mathrm{D}\}$. The buyer has a single action conditioned on all sellers: B resolves iff all sellers cooperate. Payoffs:

$$u_{S_i}(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} +P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C}\ \forall j, \\ -2G_i & \text{otherwise.} \end{cases} \tag{19}$$

The buyer's payoff is:

$$u_B(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} -\sum_{i=1}^{n} P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C}\ \forall j, \\ -\sum_{i=1}^{n} 2P_i & \text{otherwise.} \end{cases} \tag{20}$$

Theorem 5.3 (N-Party Nash Equilibrium)**.** *In Γ_N, cooperation is a weakly dominant strategy for every seller S_i at every position $i \in \{1, \ldots, n\}$, and* $(\mathrm{C}, \mathrm{C}, \ldots, \mathrm{C})$ *is the unique outcome surviving iterated elimination of weakly dominated strategies.*

Proof. Fix an arbitrary seller S_i. Consider the two cases for the remaining players' joint action profile a_{-i}.

Case 1: All other sellers cooperate and buyer resolves. Then:

$$u_{S_i}(\ldots, \mathrm{C}_i, \ldots) = +P_i, \tag{21}$$
$$u_{S_i}(\ldots, \mathrm{D}_i, \ldots) = -2G_i. \tag{22}$$

Since $P_i > 0$ and $G_i > 0$, we have $P_i > -2G_i$, i.e., $P_i + 2G_i > 0$. The cooperation payoff strictly exceeds the defection payoff.

Case 2: At least one other seller defects, or buyer defects. Then resolution does not occur regardless of S_i's action:

$$u_{S_i}(\ldots, \mathrm{C}_i, \ldots) = u_{S_i}(\ldots, \mathrm{D}_i, \ldots) = -2G_i. \tag{23}$$

Payoffs are equal.

Combining both cases, C weakly dominates D for S_i, with strict inequality in Case 1. Since the choice of i was arbitrary, this holds for all n sellers. The buyer's dominance follows from Theorem 4.3(a), generalized: the buyer's payoff under cooperation ($-\sum P_i$) strictly exceeds the payoff under defection ($-2\sum P_i$) when all sellers cooperate.

Iterated elimination: in the first round, D is eliminated for all players (weakly dominated). The unique surviving profile is $(\mathrm{C}, \ldots, \mathrm{C})$. □

Remark 5.4 (Strength of the N-Party Result). Note that the seller's dominance condition, $P_i + 2G_i > 0$, holds with a margin that *grows* with chain depth (since $G_i = \sum_{j \le i} P_j$ is monotonic). The cooperation-defection payoff gap for S_i is:

$$\Delta_i = P_i - (-2G_i) = P_i + 2G_i = P_i + 2\sum_{j=1}^{i} P_j \ge 3P_i. \tag{24}$$

The gap is at least $3P_i$ (equality only at the root where $G_i = P_i$) and grows with every upstream payment. This is the formal basis of the "geometric coordination pressure" in Theorem 5.5.

Corollary 5.5 (Geometric Coordination Pressure)**.** *The risk-to-reward ratio for seller S_i is:*

$$\rho_i = \frac{C_{s,i}}{P_i} = \frac{2G_i}{P_i} = \frac{2\sum_{j=1}^{i} P_j}{P_i}. \tag{25}$$

For equal payments ($P_j = p$ for all j), $\rho_i = 2i$, growing linearly with chain depth. For typical value chains where early stages contribute the majority of value ($P_1 \gg P_n$), ρ_n grows super-linearly.

Example 5.6. Alice orders food (\$10) from Bob, who needs delivery (\$2) from Charlie:

Party	Role	P_i	G_i	$C_{s,i}$	ρ_i
Bob	Seller 1	\$10	\$10	\$20	2.0
Charlie	Seller 2	\$2	\$12	\$24	12.0

Charlie risks \$24 to earn \$2 ($\rho = 12$). Bob risks \$20 to earn \$10 ($\rho = 2$). The deeper participant faces amplified consequences.

5.2 Atomic Resolution and Seller Coordination

Definition 5.7 (Atomic Resolution)**.** Resolution is *atomic*: the buyer must resolve all active orders in a process simultaneously, or none. There is no partial resolution.

Proposition 5.8 (Endogenous Coordination Pressure)**.** *The N-party bonding game under atomic resolution induces a* weakest-link public goods game *among sellers, where each seller's payout depends on universal cooperation.*

Proof. Consider the sellers' subgame (fixing the buyer's strategy at: resolve iff all sellers cooperate). Each S_i has payoff u_{S_i} as in (19).

Define the *externality* imposed by S_k's defection on S_i ($i \neq k$):

$$\varepsilon_{k\to i} = u_{S_i}(\text{all C}) - u_{S_i}(S_k \text{ defects}) = P_i - (-2G_i) = P_i + 2G_i. \tag{26}$$

This is strictly positive for all i, k with $i \neq k$. Every defection imposes a loss on every other seller. The total externality of S_k's defection is:

$$\mathcal{E}_k = \sum_{i\neq k} \varepsilon_{k\to i} = \sum_{i\neq k} (P_i + 2G_i). \tag{27}$$

This structure satisfies the defining properties of a *weakest-link game* [Hirshleifer, 1983]:

1. The group outcome depends on the minimum individual effort (one defection collapses all payoffs to the non-cooperative level).
2. The cooperative outcome (C, . . . , C) is the unique Pareto-optimal Nash equilibrium of the subgame.
3. The game exhibits *strategic complementarity*: S_i's marginal return to cooperation is highest when all others cooperate.

In a repeated-interaction setting, this externality structure creates endogenous peer monitoring: S_i has a direct economic interest of magnitude $P_i + 2G_i$ in ensuring S_k cooperates. This incentivizes information acquisition about co-sellers, pressure application on underperformers, and reputation-based exclusion of known defectors—all without explicit communication protocols or governance mechanisms. □

Remark 5.9 (Microfinance Analogy)*.* This structure mirrors group lending in microfinance (e.g., Grameen Bank [Grameen Bank, 1999]), where default by one member of a borrowing group causes the entire group to lose future credit access. The microfinance literature documents substantially lower default rates in group lending relative to individual lending, attributed to peer monitoring and social collateral [Ghatak, 1999].

5.3 Self-Enforcing DAG Topology

Process trees can express arbitrary directed acyclic graph (DAG) topologies. A natural question: must the contract validate parent-child relationships on-chain?

Theorem 5.10 (Self-Enforcing DAG Honesty)**.** *In any game where seller S_i chooses a reported cumulative value $G'_i \geq G_i$ (with G_i being the true value enforced by the contract's accumulator), truth-telling ($G'_i = G_i$) weakly dominates all alternatives. The dominance is strict whenever the probability of non-resolution is positive.*

Proof. The contract's monotonic accumulator enforces $G'_i \geq G_i := \Pi.G_{\text{prev}} + P_i$: any report $G'_i < G_i$ causes a revert (`CumulativeValueMismatch`), so we restrict attention to $G'_i \geq G_i$.

Let $q \in [0,1]$ be the probability that the buyer resolves the process (a function of all sellers' performance, exogenous to S_i's reporting decision). Define S_i's expected payoff as a function of the report G'_i:

$$\mathbb{E}[u_{S_i}(G'_i)] = q \cdot (\underbrace{2G'_i + P_i}_{\text{payout}} - \underbrace{2G'_i}_{\text{bond}}) + (1-q) \cdot (-2G'_i) = q \cdot P_i - (1-q) \cdot 2G'_i. \tag{28}$$

The derivative with respect to G'_i is:

$$\frac{\partial}{\partial G'_i} \mathbb{E}[u_{S_i}] = -2(1-q). \tag{29}$$

For $q < 1$ (non-zero probability of non-resolution), this derivative is strictly negative: expected utility is strictly decreasing in the reported cumulative value. The optimum is therefore at the smallest feasible value, $G'_i = G_i$.

For $q = 1$ (certain resolution), the derivative is zero and the seller is indifferent—overstating wastes capital but doesn't affect the net payoff. Truth-telling remains weakly optimal.

Since $q < 1$ in any realistic setting (the seller cannot guarantee other sellers' cooperation), truth-telling strictly dominates overstating. □

6 Implementation

6.1 Smart Contract Architecture

The protocol is implemented as a single Solidity smart contract, `FigaroCore`, inheriting OpenZeppelin's EIP-712 and ReentrancyGuard. The kernel exposes only two external functions: `commit` and `resolveProcess`.

Storage layout. Three mappings comprise the kernel state:

- `processes`: maps `bytes32` → `ProcessState` (root buyer, currency, cumulative value, active order count).
- `orderStatus`: maps `bytes32` → `uint8` (unknown, committed, resolved).
- `orderProcessId`: maps `bytes32` → `bytes32` (order membership in a process).

The kernel deliberately does not store a per-order struct. Order terms, including payment and cumulative-value snapshot, are reconstructed from the signed commitment and the emitted `OrderCommitted` event. Bond amounts (`buyerBond = 2` × `payment`, `sellerBond = 2` × `cumulativeValue`) remain deterministic functions of those values rather than separate storage variables.

Commitment protocol. Both parties sign an EIP-712 typed commitment off-chain. A single on-chain `commit` call verifies both signatures and pulls both bonds atomically. Root commitments create a new process; sub-order commitments extend an existing one by carrying the inherited `processId` and the next expected cumulative value. This eliminates the accept-reject pattern of traditional escrow protocols and prevents front-running of the counterparty's acceptance.

Identifier derivation. Process and order identifiers are content-addressed:

$$\text{processId} = H_{\text{EIP-712}}(\text{commitment}) \quad \text{for root orders,} \tag{30}$$

$$\text{orderHash} = H(\text{processId} \| \text{structHash}). \tag{31}$$

For sub-orders, `processId` is inherited from the signed commitment; the EIP-712 domain separator already binds the root identifier to chain and verifying contract. This prevents collision, replay, and cross-chain confusion without auto-incrementing counters.

Fee-on-transfer protection. The `_pullExact` function checks `balanceAfter`−`balanceBefore` = `amount`, reverting if the token applies a transfer tax. This ensures the conservation law (Theorem 3.4) cannot be broken by exotic token behavior.

6.2 Design Non-Decisions

The following features are deliberately absent:

1. **No owner or admin functions.** The contract is ownerless and has no `pause()`, `upgrade()`, or governance mechanism.
2. **No protocol fee.** Settlement distributes the full bond amount to the parties. Any associated token layer is external to the kernel; in the current runtime, seller-only settlement emission serves as a coordination Schelling point rather than a fee path (Section 9).
3. **No timeout.** An active order remains active indefinitely. Per Theorem 4.7, timeouts weaken the Nash equilibrium.
4. **No partial resolution.** Per Theorem 5.8, partial resolution would destroy the seller coordination game.
5. **No internal ledger.** Payouts are direct ERC-20 transfers, not balance increments. This eliminates withdrawal-pattern complexity and reentrancy surface.

6.3 Settlement-Anchored Emission

The kernel itself has no fee and no token dependency. However, the current runtime includes an associated coordination token (FIG) whose emission is anchored to settlement events rather than calendar time. This design is worth describing because the emission schedule is itself a mechanism design artifact with properties that follow from the protocol's structure.

Two emission paths coexist:

Direct path (on-chain). A permissionless, immutable contract (`FigEmission`) reads `FigaroCore` order state to verify resolution, then mints FIG to the seller. The rate is a discrete halving step function:

$$r_{\text{direct}}(n) = r_0 \cdot 2^{-\lfloor n/H \rfloor}, \tag{32}$$

where $r_0 = 100$ FIG per settlement, $H = 10{,}000{,}000$ settlements per halving epoch, and n is the cumulative settlement count at the time of the claim. A 600M FIG hard cap terminates emission.

Batch path (SP1 prover). The batch path uses an Euler oscillation—a decaying cosine modulation derived from Euler's identity—on top of the same decay envelope:

$$r_{\text{batch}}(n) = r_0 \cdot 2^{-n/H} \cdot \big(1 + A \cdot \cos(2\pi n/S)\big), \tag{33}$$

where $A = 0.5$ is the oscillation amplitude and $S = 2{,}000{,}000$ settlements is the season length (one full cosine cycle). The rate oscillates between 1.5× the decay envelope (peaks) and 0.5× (troughs), creating "emission seasons" whose calendar length is an emergent property of network activity.

Remark 6.1 (Zero-Integral Property). The cosine term satisfies $\int_0^S \cos(2\pi n/S)\,dn = 0$. Over each full season, the oscillation redistributes emission temporally without changing the total—the Euler path and the direct path converge in cumulative emission over long horizons.

Three design choices warrant explanation. First, the decay is settlement-count-based, not calendar-based. If adoption is slow, the high early rate persists longer (the bootstrapping subsidy is still needed); if adoption is fast, halving arrives quickly (network effects have kicked in). Second, emission is seller-only: only the party bearing the asymmetric capital commitment receives the reward. This closes wash-trading vectors (a buyer-seller pair cannot profitably cycle orders to extract FIG because each cycle requires locking $4P$ in real capital). Third, the batch oscillation creates temporal coordination value: participants who batch settlements through the SP1 prover during peak seasons earn up to 1.5× the direct rate, incentivizing the scalability path without penalizing the permissionless direct fallback.

6.4 Formal Verification

TLA$^+$ model checking. The contract is modeled in TLA$^+$ with 3 actions (`CommitRoot`, `CommitSub`, `ResolveProcess`) and 8 state variables. TLC verifies 7 invariants by exhaustive enumeration over bounded state spaces (1 buyer, 2 sellers, payments 1–3, 2 concurrent processes, 2 sub-orders each; 2,095,093 states generated, 1,521,909 distinct):

- **I1 TypeOK**: type well-formedness of all state variables.
- **I2 TokenConservation**: sum of all wallet balances plus contract balance equals total supply (conservation of value).
- **I3 ContractSolvency**: contract balance ≥ 0.
- **I4 WalletNonNegative**: all participant balances ≥ 0.
- **I5 CumulativeIntegrity**: for every process, G equals the sum of all order payments.
- **I6 ActiveCountCorrect**: stored active-order count matches the count of committed orders.
- **I7 ResolutionAlwaysPossible**: for every active process, the contract holds sufficient funds to resolve all orders.

The model abstracts EIP-712 signatures (assumed correct given valid commitments), ERC-20 mechanics (modeled as integer balances), and timing (deadlines are orthogonal to the bonding equilibrium).

Property-based fuzzing (Echidna). Seven property invariants are checked after every call sequence (50,000 sequences of length 30, totaling ∼43,000 individual calls):

- **E1 Solvency**: token balance $\geq$ sum of outstanding bonds.
- **E2 ActiveCountConsistent**: stored count matches actual.

E3 CumulativeAccounting: accumulator equals sum of payments.

E4 StateMonotonicity: orders never transition backwards.

E5 BuyerDominance: non-buyer resolve always reverts.

E6 AtomicResolution: incomplete order list always reverts.

E7 NoSelfDeal: $B = S$ always reverts.

The Echidna fuzzer uses HEVM cheatcodes for EIP-712 signing with known private keys, exercising the full commitment-to-resolution path.

Unit testing. 236 Foundry unit tests cover the contract across 15 test files, including EIP-712 signature verification, cumulative value enforcement, permit integration, event emission, revert branch coverage, batch verifier parity, FIG emission mechanics, and safety hardening. An additional 166 tests cover the TypeScript SDK.

7 Security Analysis

7.1 Threat Model

We assume rational adversaries who maximize economic payoff. We consider four attack vectors.

Griefing (buyer refuses to resolve). The attacker's cost is $2P$ (locked bond) plus opportunity cost on that capital plus permanent on-chain reputation damage. The attacker's benefit is zero (no economic gain from non-resolution). For rational agents, the attack is strictly dominated by cooperation. For irrational agents (spite exceeds economic loss), Layer 3 provides deterrence: the immutable on-chain record—including the order's commitment timestamp, the buyer's address, and the fact that no resolution has been called—serves as evidence in off-chain legal proceedings.

Sybil (fake orders to manipulate state). Each order requires real bond deposits: $2(G_i+P_i)$ total locked per order. At scale, Sybil attacks are capital-intensive with no mechanism for extracting the invested capital (there is no secondary market, no yield, and no governance influence associated with locked bonds).

Front-running. Order identifiers are content-addressed from the signed commitment. An attacker who observes a pending `commit` transaction cannot create a conflicting order for the same process (different signatures produce different hashes). No extractable MEV exists.

Cumulative value manipulation. Per Theorem 5.10, overstating cumulative value is strictly dominated (greater capital at risk, same payoff). Understating is prevented by the contract's monotonic accumulator check.

7.2 Liveness

Theorem 7.1 (Liveness Under Rationality). *Let B be a buyer with time-discounted utility $U_B = \sum_{t=0}^{\infty} \delta^t \cdot u_t$ where $\delta \in (0,1)$ is the discount factor and u_t is the per-period payoff. If all sellers have cooperated (the buyer has received the agreed goods), then the buyer resolves in finite time.*

Proof. Let $\mathcal{B} = \sum_{i=1}^{n} 2P_i$ be the buyer's total locked capital. Define the buyer's per-period payoff in each scenario:

- **Resolve at time** t^*: Payoff at t^* is $\sum_i \pi_{b,i} = \sum_i P_i$ (recovered capital). Payoff for $t < t^*$ is 0 (capital locked, goods received). Total: $\delta^{t^*} \sum_i P_i$.
- **Never resolve**: Payoff at every t is 0 (capital locked forever, goods received but excess capital not recovered). Total: 0.

The buyer prefers resolving at t^* over never resolving iff:

$$\delta^{t^*} \sum_i P_i > 0, \tag{34}$$

which holds for all finite t^* since $\delta > 0$ and $P_i > 0$.

The buyer prefers resolving sooner (lower t^*) since $\delta^{t_1} > \delta^{t_2}$ for $t_1 < t_2$ and $\delta < 1$. Therefore the optimal strategy is $t^* = 0$ (resolve immediately).

A buyer who does not resolve at $t = 0$ forgoes $\delta^0 \sum_i P_i = \sum_i P_i > 0$ in present value. Since this exceeds the benefit of non-resolution (which is 0 for a rational agent with no taste for spite), indefinite non-resolution is strictly dominated. □

Remark 7.2 (Limitations). The result assumes: (i) the buyer has received satisfactory performance from all sellers, (ii) the buyer has a positive discount rate ($\delta < 1$), and (iii) the buyer does not derive positive utility from harming sellers (no spite). Condition (iii) is the hardest to guarantee. When it fails—an adversarial buyer who values punishment above wealth—the mechanism cannot guarantee liveness by economic incentives alone. This is the residual case addressed by Layer 3 (legal deterrence via immutable on-chain evidence).

7.3 The Three-Layer Enforcement Architecture

Table 2: Defense-in-depth: three enforcement layers.

Layer	Mechanism	Coverage	Failure Mode
1. Bonding	Asymmetric collateral	>99% of orders	Irrational actors
2. Coordination	Atomic resolution	Multi-seller failures	Uncoordinated sellers
3. Legal	Immutable evidence	Adversarial actors	Jurisdictional limits

Layer 1: Economic. Theorems 4.3 and 5.3 establish that cooperation is strictly dominant for all rational participants. This layer handles the vast majority of transactions without any interaction beyond commit-and-resolve.

Layer 2: Social. Theorem 5.8 shows that atomic resolution creates endogenous peer pressure. In multi-seller processes, honest sellers have direct economic incentive to monitor and pressure underperforming co-sellers. This layer requires no explicit communication protocol; the incentive structure is sufficient.

Layer 3: Legal. For the residual case of truly irrational or adversarial actors, the on-chain record provides tamper-proof evidence. Every commitment, every bond deposit, every resolution (or lack thereof) is recorded with block timestamps. This evidence can be presented in any off-chain legal forum. Importantly, the protocol does *not* attempt to perform dispute resolution on-chain; it provides the *evidence* for off-chain systems that already exist.

8 Implications for Organizational Theory

8.1 From Firms to Transaction-Scoped Institutions

Coase's [1937] central insight is that firms exist because market transaction costs—searching for partners, negotiating terms, enforcing agreements—sometimes exceed the cost of hierarchical management. Williamson [1985] refined this into Transaction Cost Economics (TCE), identifying three conditions under which hierarchical governance dominates market exchange: high asset specificity, high uncertainty, and high frequency.

Asymmetric bonding changes this calculus. Consider the three transaction costs separately:

Search costs. Public on-chain signals (settlement history, accepted-token lists, geolocation proofs) allow autonomous discovery through shared coordination graphs. Search cost approaches zero for agents that can read blockchain state.

Negotiation costs. EIP-712 off-chain commitment building allows parties to iterate on terms (payment, agreement hash, token, topology) without gas costs. The negotiation is bilateral and direct; counterparties can discover and iterate through shared graph visibility rather than through a standing intermediary's matching function.

Enforcement costs. This is the decisive variable. In the pre-Figaro world, enforcement requires courts (slow, expensive, jurisdiction-dependent), arbitrators (trusted third parties), or standing intermediaries that internalize coordination and charge for it. FIGARO prices enforcement at exactly $2\times$ the transaction value—a fixed, predictable, jurisdiction-independent cost requiring no third party.

Proposition 8.1 (Coasean Threshold Shift)**.** *For any transaction where $2P$ (the capital lockup cost) is less than the firm's coordination overhead for the same transaction, the rational organizational choice is market exchange via asymmetric bonding rather than hierarchical integration.*

The implication is structural: the boundary of the firm shifts *inward*, but the stronger implication is organizational rather than merely contractual. When enforcement is priced directly in bonded capital, the standing firm ceases to be the default container of production. What emerges instead is a *transaction-scoped institution*: a temporary assembly of directly bonded counterparties that forms around a process, coordinates for the duration of that process, and dissolves at settlement. Functions previously internalized for trust reasons—procurement, quality assurance, fulfillment verification—become candidates for sovereign coordination between independent value-adders. The "restaurant" becomes a coordination label rather than a unitary firm: a cook, a kitchen operator, an ingredient sourcer, and a courier each bond and settle independently within the same process tree.

Each seller node in a process tree corresponds to what we term a *key component*—the irreducible asset or capability that provides competitive advantage in a specific market [Daliana, 2013]. In the firm model, key components are embedded inside organizational hierarchies: the cook's skill, the kitchen's equipment, the courier's vehicle are all subsumed under the firm's legal identity. In the FIGARO model, each key component is represented directly by a seller wallet address. The bonded process tree does not merely flatten the firm—it makes each value-adding capability individually addressable, individually compensated, and individually accountable. The productive asset *is* the identity.

Nothing in the mechanism requires that a node be a natural person or a corporation. Any actor capable of controlling an address and bearing capital risk—a human, a software agent, a machine operator, or a treasury—can occupy a node in the process tree. The accountable unit is the bonded counterparty, not the firm.

8.2 Token Denomination as Coordination Signal

In Figaro, the choice of denomination token is itself a coordination signal. The protocol is token-agnostic: any ERC-20 can serve as the bond and payment currency. This means:

- A stablecoin signals legal-system alignment.
- A DAO governance token signals community membership.
- A commodity-backed token signals value anchoring.
- Accepting a specific token *is* the seller's identity declaration—an accepted-token list is a brand.

Settlement velocity in a given token replaces traditional reputation metrics (star ratings, review scores). The on-chain record of resolved orders denominated in token τ is a public, tamper-proof signal of the seller's reliability within the τ economy. No platform needs to aggregate, curate, or attest this signal; it is a natural byproduct of protocol usage.

8.3 Cultural Hegemony and the Visibility of Coordination

Gramsci [1971] observed that the most durable forms of power are those embedded so deeply in common sense that they become invisible. He called this *cultural hegemony*: the dominant class rules not only through coercion but by making its arrangements appear natural, inevitable, and beyond question. The concept applies directly to contemporary coordination infrastructure.

The platform economy illustrates Gramscian hegemony precisely. The proposition that two strangers need a platform to transact safely feels like common sense. It is not. It is a contingent arrangement that arose because no economic primitive existed for safe direct exchange between strangers. In Gramscian terms, the platform has achieved hegemonic status: its necessity is no longer argued for—it is assumed.

Similarly, the proposition that production requires firms (the Coasean assumption examined in Theorem 8.1) and that cross-border trade requires corridors controlled by state or alliance actors (ports, railways, digital infrastructure) have both achieved the same hegemonic status. These are presented as structural necessities rather than contingent design choices.

Asymmetric bonding makes the coordination function *visible* by separating it from the entities that currently perform it. The platform's coordination function (matching, enforcement, dispute resolution) becomes a public smart contract. The firm's coordination function (search, negotiation, enforcement) becomes three reducible costs. The corridor's coordination function (settlement, standards, access control) becomes a bilateral mechanism. In each case, the entity does not need to be *replaced*—it needs to be *seen* as a design choice rather than a fact of nature. Once the coordination function is visible and separable, the architecture that makes the entity structurally unnecessary follows.

This analysis explains a practical difficulty in communicating the protocol's implications. Proposing to remove structures that most people do not know are there is qualitatively different from proposing a better version of the same structure. The challenge is not engineering. It is making existing infrastructure legible as architecture rather than gravity.

9 Conclusion and Future Work

We have presented Figaro, a smart-contract protocol that achieves self-enforcing multi-party coordination through asymmetric bonding. The mechanism is minimal (two external functions, no owner, no fee, no escape hatches) and the game-theoretic properties are strong (cooperation is strictly dominant at every position in an N-party process tree).

The implementation is formally verified through TLA^{+} model checking (7 invariants, 2M+ states), property-based fuzzing, and 180 Foundry unit tests. The three-layer enforcement architecture—economic, social, legal—provides defense-in-depth without on-chain governance or dispute resolution.

Any associated token layer remains external to the kernel. In the current runtime, seller-only settlement emission serves as a coordination Schelling point; it is not required for bonding, settlement, or participation.

Mechanism composition. The kernel's bonding equilibrium is a fixed point. The question—under what conditions does an external mechanism composed with the kernel preserve the Nash equilibrium?—is answered by the implementation's convergence to the *coordinator pattern.*

An external mechanism preserves the equilibrium if it satisfies four constraints: (i) read-only interaction with core state (`staticcall` or event consumption), (ii) no impersonation of buyer or seller in core functions, (iii) no modification to locked bonds, and (iv) no bypass of buyer-only resolution.

The reference implementation contains three proof cases. First, a Dutch auction performs pure price discovery and role allocation with zero token custody: the auction determines who may become the seller, but the actual bond is still posted only through `commit`. Second, the attestation coordinator appends schema-typed lifecycle and disclosure events with role gating, but never touches locked capital or resolution authority. Third, the schema registry and operator registry anchor shared reference semantics and participant metadata as event-first coordination layers; they are visible to agents and frontends, but cannot alter the payoff matrix or bypass buyer-only resolution. All compose with the kernel without weakening Theorem 4.3.

Mechanism composition is therefore a layer above the kernel. The kernel does not know or constrain what is composed around it. Declarative institution assemblies specify which mechanisms compose for a given workflow, with explicit risk classification (read-only, low-risk coordinator, medium-risk extension, high-risk economic). The coordinator pattern guarantees that read-only and event-only mechanisms preserve the equilibrium by construction.

Broader implication. The Coasean threshold is not a conjecture—it is observable. Contemporary coordination stacks often extract 15–30% of transaction value as coordination rent (Uber 25%, DoorDash 30%, Airbnb 14%, Amazon Marketplace 15%). Credit card networks layer 2–3% interchange plus double-digit annual interest on revolving balances. Financial markets intermediate trillions in notional value against a fraction of that in real economic exchange. Asymmetric bonding prices the cost of trust at $2\times$ the transaction value in temporarily locked (not spent) capital, with zero recurring extraction. For any economic activity where the coordination rent exceeds the opportunity cost of locked capital, the standing firm—as Coase conceived it—is no longer the uniquely efficient unit of economic organization. The stronger consequence is the emergence of transaction-scoped institutions: temporary assemblies of directly bonded contributors that form around a process and dissolve at settlement. The protocol does not abolish coordination; it changes the unit of coordination from the standing firm to the bonded process.

Agent-native design. The protocol is designed for autonomous agent participation as a first-class concern. EIP-712 typed commitments allow agents to build, verify, and sign structured data using standard libraries. Event-sourced state reconstruction means agents can derive the complete process graph from on-chain logs without subgraph dependencies. The companion SDK exposes a typed action proposer (`proposeActions`) that takes a process state and an address, returning the set of available actions; and a dual-mode execution queue (`ActionQueue`) that supports both human-in-the-loop approval and fully autonomous submission. Nothing in the mechanism distinguishes a human signer from an autonomous agent: the bonding equilibrium holds for any rational actor capable of controlling an address and bearing capital risk.

Execution environments. Future work includes verifying the kernel in alternative execution environments, including the ongoing Cairo/Starknet port. The key question is not only throughput, but which execution environments preserve atomic resolution, buyer dominance, and evidence visibility most cleanly without fragmenting a single process across multiple settlement domains.

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